

Towards Embodiment-Agnostic Human-to-Robot Motion Retargeting: A Position on Open Challenges

Anonymous Author(s)

Abstract

Enabling robots to learn from human demonstration is one of the most promising directions in robot learning, and motion retargeting, transferring human motion captured from wearable sensors or video onto a robot with a fundamentally different body, is a core component of it. Despite a decade of progress, most retargeting methods are built for a single robot platform and require substantial manual re-engineering to support a new one. This paper surveys the current landscape of human-to-robot motion retargeting and organises it into four families: geometric and kinematic mapping, optimisation-based retargeting, learning-based retargeting, and physics-aware imitation. From this survey we identify three recurring open problems, physics feasibility, morphology dependence, and input-quality sensitivity, and we argue for a shift toward embodiment-agnostic pipelines: systems that accept arbitrary human motion input and produce physically valid trajectories for arbitrary target robots without per-robot tuning. We close with a concrete research question and a plan for evaluating such a pipeline.

1. Introduction

Robots that can learn directly from human demonstration could remove much of the expert programming that currently limits their deployment in manufacturing, healthcare, and service settings. The underlying idea is simple: record what a person does using sensors or cameras, and transfer that motion to a robot. The difficulty lies entirely in the transfer. Human and robot bodies differ in their joints, link lengths, and degrees of freedom, so a direct copy of joint angles almost never produces a valid robot motion. Bridging this gap is the problem of motion retargeting.

The field has advanced rapidly in recent years, but it has matured narrowly. Methods are typically demon-

strated on one robot, a Pepper, a NAO, a single dexterous hand, or one full-sized humanoid, and the mapping, cost weights, or training data are tailored to that platform. Moving to a new robot usually means redoing much of the engineering. We argue that this per-robot specialisation, rather than any single technical bottleneck, is now the principal obstacle to scaling demonstration-based robot learning.

This paper makes three main contributions. First, we provide a structured taxonomy of retargeting methods grounded in fifteen representative works spanning robotics and computer graphics, summarised in a comparison table. Second, we identify three open problems that cut across all of them. Third, we take the position that the next advance must come from embodiment-agnostic retargeting, and we frame a concrete research question and evaluation plan to pursue it.

2. Background

2.1. What is retargeting?

Retargeting is a way of transferring motion from one source to another, in our case the source is a human and the target is a robot, while preserving the intent of the motion and respecting the robot's constraints. The input may come from a wearable inertial suit, optical motion capture, or video-based pose estimation. The output is a sequence of robot joint-angle trajectories that should be both kinematically valid and physically executable.

2.2. The embodiment gap

The central obstacle is the *embodiment gap* between the human and the robot. It appears in three forms. There is a *structural mismatch*, because the human's ball-and-socket joints rarely correspond cleanly to a robot's revolute joints. There is a *degree-of-freedom mismatch*, because a robot segment may have fewer or

more controllable axes than the human limb it imitates. And there is a *feasibility gap*, because a motion that is comfortable for a person may violate the robot’s joint limits, cause self-collision, or be dynamically impossible to execute.

2.3. Defining embodiment-agnostic retargeting

We use the term *embodiment-agnostic* to describe a retargeting pipeline that satisfies three properties. First, it accepts a robot’s kinematic and dynamic description (e.g. a URDF file) as runtime input rather than encoding robot-specific parameters during training or design. Second, it produces trajectories that are physically feasible for that robot without per-robot cost tuning or manual correspondence design. Third, it can be applied to robots of structurally different classes, such as a serial robot arm, a humanoid torso, or a dexterous hand, without architectural changes to the pipeline. This is deliberately a stronger requirement than multi-robot generalisation within a single structural class (e.g. humanoid-to-humanoid), which several existing methods already achieve.

3. Survey of Existing Approaches

3.1. Geometric and kinematic mapping

The most direct approaches define an explicit geometric correspondence between human and robot. Khalil et al.’s Pepper pipeline scales human limb vectors proportionally to the robot’s links, which is cheap and runs from ordinary video, but is sensitive to the noise and depth ambiguity of single-camera pose estimation, limiting it to expressive, low-precision motion [2]. Darvish et al. generalise this idea with whole-body geometric retargeting, which formulates the correspondence as inverse kinematics over the full robot model and is explicitly designed for scalability: different robots can be teleoperated by different users with only minimal changes to the system [4]. This scalability goal is precisely the direction this paper looks into, although the method still depends on a hand-designed correspondence per robot.

In real-time teleoperation, the same kinematic-mapping idea is tuned for responsiveness and naturalness rather than offline data generation. Mohan and Kuchenbecker’s OCRA (Optimization-based Customizable Retargeting Algorithm) maps an operator’s upper-body motion from a commercial inertial suit onto a NAO humanoid in real time, deliberately exploiting the redundant degrees of freedom that end-effector-only methods discard, so that the robot’s whole arm, not just

its hand, moves in a human-like way [3]. Such systems prioritise perceptual quality and are well suited to social robotics, but they are not intended to generate training data for autonomous policies.

3.2. Optimisation-based retargeting

Optimisation-based methods cast retargeting as a constrained problem: minimise the discrepancy between human and robot pose while satisfying joint limits and avoiding collisions. Wang et al. follow this template for a NAO humanoid, acquiring human motion from inertial measurement units and combining kinematic modelling, inverse kinematics, constraint enforcement, and a final difference-optimisation step to produce balanced, human-like upper-limb motion [5]. The approach achieves a good kinematic match but is computationally heavy and requires its cost terms to be tuned for the target robot.

The same paradigm scales to dexterous hands. DexPilot teleoperates a 23-degree-of-actuation hand-arm system from nothing more than four depth cameras observing a bare human hand: a nonlinear optimiser aligns a set of wrist-to-fingertip and inter-finger vectors between the human and robot hands via forward kinematics, producing fine manipulation behaviour without gloves or markers [6]. DexPilot shows both the power and the cost of optimisation-based retargeting; it is accurate and general within a hand morphology, but the vector correspondences and objective are made specifically for that hand.

3.3. Learning-based retargeting

Learning-based methods replace manual, hand-designed rules with mapping systems learned directly from data. The goal is to automatically teach the system what movements are physically possible for a robot, making it easier to adapt to new robot designs. The main difference among these methods is how much matched human-robot data they need to train.

Zhang et al. map human movements to robots by using a decoder network that searches a compressed latent space to find the best-matching robot pose. By modelling both humans and robots as connected structural graphs, they successfully transferred complex, two-armed sign language to three different robots (YuMi, NAO, and Pepper) [7]. To address the problem of scarce training data, Choi et al. introduced a self-supervised method that automatically generates its own matching data. Crucially, they encoded mathematical boundaries in their data mapping to guarantee that every generated movement is within safe position limits

and completely free of self-collisions [8].

Kim and Lee’s C-3PO system uses deep reinforcement learning in a continuous three-phase loop that refines movements with a human evaluator in the loop, demonstrating control of NAO, Pepper, and Baxter robots with a single framework [9]. Yan et al. created ImitationNet, which completely removes the need for paired data by building a shared space where human and robot movements are compared using a universal similarity metric, allowing the robot to be controlled by text, regular video, or simple poses [10]. The authors later extended this to dexterous hands using standard video [11]. These learning-based approaches complement foundational work on robot learning from demonstration, which has established the broader framework for encoding task-level behaviour from human examples [16, 17].

3.4. Physics-aware imitation and cross-morphology retargeting

This approach treats retargeting not just as a problem of matching body positions, but as the first step in physical control. It acknowledges that a mathematically perfect movement is useless if a physical robot cannot actually execute it without falling over or breaking.

DeepMimic is the foundational work in this area, using reinforcement learning to imitate reference motion clips and demonstrating that training a system with physics in mind produces stable, natural movements [13]. HumanMimic applies this concept to a real, full-sized humanoid robot (JAXON), pairing a simplified skeleton-mapping step with an adversarial scoring system that forces the robot to stay within safe, natural movement boundaries. It even allows the robot to naturally transition between gaits not explicitly shown in the training data [14].

The research that comes closest to handling any type of robot body originates in the computer graphics field. ACE (Adversarial Correspondence Embedding) retargets human movements onto characters with structurally different bodies, such as a four-legged creature with a robotic arm, a crab, and a wheeled robot with an extending arm [15]. ACE proves that a single framework can bridge completely different body shapes, a capability that robotics still struggles with, but its main limitation is a focus on visual realism rather than the physical-contact constraints required by actual hardware. Recent work on morphology-aware imitation learning has begun to address this gap by conditioning policies on robot morphology descriptions at runtime, allowing a single learned policy to transfer across robots with different limb counts and proportions [18].

3.5. Comparison of surveyed methods

Table 1 summarises the surveyed methods along five dimensions relevant to embodiment-agnostic retargeting.

The table makes the core gap visible: no method simultaneously achieves multi-class generalisation, physics feasibility, and freedom from per-robot tuning. ACE comes closest on the first two criteria but is limited to animated characters; Choi et al. and ImitationNet eliminate per-robot tuning but stay within a single structural class and do not enforce physical feasibility.

4. Open Problems

Three problems recur across these four families. The first is **physics feasibility**. Most geometric, optimisation-based, and learning-based methods produce trajectories that are geometrically valid but dynamically impossible to execute. A trajectory that exceeds a robot’s joint torque limits or destabilises its balance is functionally useless. Only physics-aware methods like DeepMimic and HumanMimic confront this issue directly.

The second is **morphology dependence**. The majority of surveyed methods are restricted to a single hardware platform. The few frameworks that handle multiple robots are confined to a single structural class; for instance, C-3PO, ImitationNet, and Zhang et al. only generalise across humanoids with highly similar structures. Conversely, while ACE successfully spans completely different body shapes, it is limited to animated characters. No existing method provides a universal framework that adapts across distinct physical robot structures without manual re-engineering.

The third is **input-quality sensitivity**. Video-based approaches degrade rapidly under poor lighting or physical occlusion. In contrast, wearable motion-capture suits are highly robust to environmental conditions but introduce heavy calibration burdens. No current pipeline is capable of robustly integrating both input modalities.

5. Position: The Case for Embodiment-Agnostic Pipelines

Because single-robot tracking is largely solved, we argue that the next breakthrough lies in abstraction: a unified pipeline that maps any human input to any target robot morphology without robot-specific tuning.

The missing link is a *robot-independent human motion representation* combined with a *feasibility layer*

Table 1. Comparison of surveyed human-to-robot motion retargeting methods. *Multi-class* = generalises across structurally different robot families. *Phys. feas.*: None = not addressed; Constr. = constraint/IK optimisation; Sim. = physics simulation or RL.

Method	Input	Target robot(s)	Multi-class?	Phys. feas.	Per-robot tuning?
Khalil et al. [2]	Video (monocular)	Pepper (humanoid)	No	None	Yes
Darvish et al. [4]	Wearable suit	Multiple humanoids	No	Constr.	Yes (per robot)
Mohan & Kuchenbecker [3]	Wearable suit	NAO (humanoid)	No	Constr.	Yes
Wang et al. [5]	IMU suit	NAO (humanoid)	No	Constr.	Yes
DexPilot [6]	Depth cameras	LEAP/Allegro hand-arm	No	Constr.	Yes
Zhang et al. [7]	Video / MoCap	YuMi, NAO, Pepper	No	None	Partial
Choi et al. [8]	MoCap	Humanoid (generic)	No	Constr.	No
C-3PO [9]	Video	NAO, Pepper, Baxter	No	None	Partial
ImitationNet [10]	Video / text	Humanoids (multiple)	No	None	No
Yan & Lee [11]	Video	Dexterous hands	No	None	No
Qin et al. [12]	Single camera	Dexterous hand	No	None	Yes
DeepMimic [13]	MoCap	Simulated character	No	Sim.	Yes
HumanMimic [14]	MoCap	JAXON humanoid	No	Sim.	Yes
ACE [15]	MoCap	Animated cross-morph chars	Yes	Sim.	No
Gupta et al. [18]	MoCap / state	Multiple morph. robots	Yes	Sim.	Partial

that automatically adapts to a target robot’s constraints. By robot-independent representation we mean a body-relative kinematic encoding, such as normalised joint directions, velocity fields, or contact events, that captures the intent of a motion without committing to any particular skeleton structure. This representation would be consumed by a feasibility layer that reads the target robot’s URDF description at runtime and applies the appropriate joint-limit and collision-avoidance constraints, rather than having them baked in at training time.

This leads to our research question: *Can a single retargeting pipeline generate physically feasible trajectories across multiple robot morphologies, from wearable-suit input, without per-robot re-engineering?*

We propose evaluating this by driving both a UR5 and a Franka Emika Panda arm in simulation using motion data from wearable suits. These two arms were selected as a deliberately controlled first test: despite both being serial manipulators, they differ meaningfully in their kinematic structure. The UR5 has a spherical wrist with six revolute joints and a long reach, while the Franka has a different joint-angle layout and a compliance-oriented design. This makes them a non-trivial pair within a well-defined class before scaling to more radically different platforms such as humanoids or dexterous hands. Feasibility will be enforced via the physics simulator and benchmarked against a classical

inverse-kinematics baseline. Success across these two arms with zero custom tuning would provide the first empirical evidence for a truly embodiment-agnostic pipeline; failure would reveal precisely where the representation or feasibility layer breaks down.

6. Conclusion and Future Work

This paper surveyed geometric, optimisation, learning, and physics-aware motion retargeting, identifying physical feasibility, morphology dependence, and input sensitivity as core open challenges, and summarised key distinctions between methods in a structured comparison table. While current research favours per-robot specialisation, we argue that overcoming this fragmentation is the field’s most valuable next step. We defined embodiment-agnostic retargeting precisely: a pipeline that takes a robot’s kinematic description as runtime input, applies a robot-independent human motion representation, and enforces physical feasibility automatically, without architectural changes or per-robot tuning across structurally different robot classes.

Future work will develop and evaluate a pipeline of this kind using wearable-suit data to control simulated UR5 and Franka arms, benchmarked against classical inverse kinematics. A natural subsequent step would extend evaluation to more dissimilar platforms such as humanoids and dexterous hands, and investigate robust

integration of video-based pose estimation as a complementary input modality.

References

- [1] K. Darvish, L. Penco, J. Ramos, R. Cisneros, J. Pratt, E. Yoshida, S. Ivaldi, and D. Pucci, “Teleoperation of Humanoid Robots: A Survey,” *IEEE Transactions on Robotics*, vol. 39, no. 3, pp. 1706–1727, 2023.
- [2] H. Khalil, E. Coronado, and G. Venture, “Human Motion Retargeting to Pepper Humanoid Robot from Un-calibrated Videos Using Human Pose Estimation,” in *Proc. 30th IEEE Int. Conf. on Robot and Human Interactive Communication (RO-MAN)*, 2021.
- [3] M. Mohan and K. J. Kuchenbecker, “My Robot, My Motion: Expressive Real-Time Teleoperation,” in *Proc. 20th ACM/IEEE Int. Conf. on Human-Robot Interaction (HRI)*, 2025.
- [4] K. Darvish, Y. Tirupachuri, G. Romualdi, L. Rapetti, D. Ferigo, F. J. A. Chavez, and D. Pucci, “Whole-Body Geometric Retargeting for Humanoid Robots,” in *Proc. IEEE-RAS 19th Int. Conf. on Humanoid Robots (Humanoids)*, 2019.
- [5] W. Wang, W. Yu, X. Qin, H. Wang, J. Hong, and Y. Feng, “Human Motion Tracking Control for Humanoid Robot Based on the Optimized Motion Retargeting,” 2022.
- [6] A. Handa, K. Van Wyk, W. Yang, J. Liang, Y.-W. Chao, Q. Wan, S. Birchfield, N. Ratliff, and D. Fox, “DexPilot: Vision-Based Teleoperation of Dexterous Robotic Hand-Arm System,” in *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2020, pp. 9164–9170.
- [7] H. Zhang, W. Li, J. Liu, Z. Chen, Y. Cui, Y. Wang, and R. Xiong, “Kinematic Motion Retargeting via Neural Latent Optimization for Learning Sign Language,” *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 4582–4589, 2022.
- [8] S. Choi, M. J. Song, H. Ahn, and J. Kim, “Self-Supervised Motion Retargeting with Safety Guarantee,” in *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2021, pp. 8097–8103.
- [9] T. Kim and J.-H. Lee, “C-3PO: Cyclic-Three-Phase Optimization for Human-Robot Motion Retargeting based on Reinforcement Learning,” in *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2020.
- [10] Y. Yan, E. Valls Mascaro, and D. Lee, “ImitationNet: Unsupervised Human-to-Robot Motion Retargeting via Shared Latent Space,” in *Proc. IEEE-RAS 22nd Int. Conf. on Humanoid Robots (Humanoids)*, 2023.
- [11] Y. Yan and D. Lee, “Learning Dexterous Robot Hand Control by Imitating Human Hands,” in *Proc. 22nd Int. Conf. on Ubiquitous Robots (UR)*, 2025.
- [12] Y. Qin, H. Su, and X. Wang, “From One Hand to Multiple Hands: Imitation Learning for Dexterous Manipulation from Single-Camera Teleoperation,” *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 10873–10881, 2022.
- [13] X. B. Peng, P. Abbeel, S. Levine, and M. van de Panne, “DeepMimic: Example-Guided Deep Reinforcement Learning of Physics-Based Character Skills,” *ACM Transactions on Graphics (SIGGRAPH)*, vol. 37, no. 4, 2018.
- [14] A. Tang, T. Hiraoka, N. Hiraoka, F. Shi, K. Kawaharazuka, K. Kojima, K. Okada, and M. Inaba, “HumanMimic: Learning Natural Locomotion and Transitions for Humanoid Robot via Wasserstein Adversarial Imitation,” in *Proc. IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2024.
- [15] T. Li, J. Won, A. Clegg, J. Kim, A. Rai, and S. Ha, “ACE: Adversarial Correspondence Embedding for Cross Morphology Motion Retargeting from Human to Nonhuman Characters,” in *ACM SIGGRAPH Asia 2023 Conference Papers*, 2023.
- [16] B. D. Argall, S. Chernova, M. Veloso, and B. Browning, “A Survey of Robot Learning from Demonstration,” *Robotics and Autonomous Systems*, vol. 57, no. 5, pp. 469–483, 2009.
- [17] H. Ravichandar, A. S. Polydoros, S. Chernova, and A. Billard, “Recent Advances in Robot Learning from Demonstration,” *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 3, pp. 297–330, 2020.
- [18] A. Gupta, L. Fan, S. Ganguli, and L. Fei-Fei, “Embodied Intelligence via Learning and Evolution,” *Nature Communications*, vol. 12, no. 1, p. 5721, 2021.